

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch			B.Tech Artificial Intelligence & Data Science	Semester V
Subject Name	Code	&	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are:

- To study the evolution of deep Convolutional Neural Network architectures.
- To compare important CNN architectures including LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- To understand the concept of transfer learning.
- To implement transfer learning using a pretrained CNN.
- To fine tune selected layers of a pretrained CNN.
- To evaluate the performance of the model using standard classification metrics.

2. Learning Outcomes

After completing this experiment, the student will be able to:

1. Explain the evolution of modern CNN architectures.
2. Differentiate between LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
3. Understand the principles of transfer learning.
4. Implement transfer learning using a pretrained CNN.
5. Fine tune selected pretrained convolutional layers.
6. Evaluate image classification models using Accuracy, Precision, Recall and F1-score.

3. Introduction

Deep Convolutional Neural Networks (CNNs) have become fundamental models for image classification and computer vision tasks. CNNs learn hierarchical representations of images through convolutional filters, activation functions and pooling operations.

The development of CNN architectures has progressed from relatively small networks such as LeNet-5 to deeper and more sophisticated architectures such as AlexNet, VGG16, GoogleNet and ResNet. These architectures introduced important concepts such as deeper convolutional networks, parallel convolutional branches and residual connections.

Training deep neural networks from scratch can require large datasets, considerable computational resources and long training times. Transfer learning addresses this problem by reusing knowledge learned by a model on a large dataset.

In this experiment, MobileNetV2 pretrained on ImageNet is used for classification of CIFAR-10 images. The pretrained convolutional base is initially frozen and a new classification head is trained. Selected layers are subsequently unfrozen and fine tuned using a smaller learning rate.

4. Evolution of CNN Architectures

The evolution of CNN architectures has focused on increasing feature learning capability while improving computational efficiency and optimization.

- **LeNet-5:** One of the earliest successful CNN architectures, mainly designed for handwritten digit recognition.
- **AlexNet:** Demonstrated the effectiveness of deep CNNs on large-scale image classification and popularized the use of ReLU activations and GPU-based training.
- **VGG16:** Used a deeper architecture based mainly on small 3×3 convolutional filters.
- **GoogLeNet:** Introduced the Inception module, allowing multiple convolutional operations with different receptive fields to operate in parallel.
- **ResNet:** Introduced residual connections to make training very deep neural networks easier.

5. LeNet-5

LeNet-5 is an early convolutional neural network architecture developed for handwritten digit recognition.

Its architecture consists of convolutional and pooling layers followed by fully connected layers.

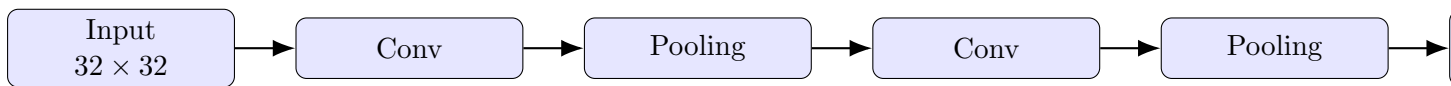


Figure 1: Simplified LeNet-5 Architecture

LeNet demonstrated that convolutional operations could automatically learn useful spatial features from images and became a foundation for later CNN architectures.

6. AlexNet

AlexNet was a major milestone in deep learning for image classification. It achieved strong performance on the ImageNet Large Scale Visual Recognition Challenge.

Important features of AlexNet include:

- ReLU activation functions.
- GPU-based training.
- Multiple convolutional layers.
- Dropout for reducing overfitting.
- Data augmentation.

The use of ReLU significantly improved optimization compared with traditional sigmoid or tanh activations.

7. VGG16

VGG16 is a deep CNN architecture containing 16 weight layers. It uses small 3×3 convolutional filters throughout most of the network.

The architecture can be summarized as:

Convolution $\rightarrow$ ReLU $\rightarrow$ Pooling $\rightarrow$ Fully Connected

The repeated use of small convolutional filters allows VGG16 to learn hierarchical visual representations while maintaining a relatively simple architecture.

8. GoogleNet and ResNet

8.1 GoogleNet

GoogleNet introduced the Inception architecture. Instead of relying on a single convolutional filter size, the Inception module applies multiple operations in parallel.

- 1×1 convolution
- 3×3 convolution
- 5×5 convolution
- Pooling

The outputs are concatenated to form a richer feature representation.

8.2 ResNet

ResNet introduced residual learning using shortcut connections.

A residual block can be represented as:

$$y = F(x) + x$$

where $F(x)$ represents the learned residual mapping and x is the identity shortcut.

Residual connections make it easier to train very deep networks and help reduce the degradation problem.

9. Comparison of CNN Architectures

Architecture	Main Contribution	Characteristic	Limitation
LeNet-5	Early CNN architecture	Small convolutional network	Limited capacity for complex datasets
AlexNet	Deep CNN and ReLU	GPU training and dropout	Large computational requirements
VGG16	Deep network with 3×3 filters	Simple and uniform architecture	Large number of parameters
GoogleNet	Inception modules	Parallel multi-scale feature extraction	More complex architecture
ResNet	Residual connections	Very deep networks	Higher computational complexity

Table 1: Comparison of Major CNN Architectures

10. Transfer Learning

Transfer learning is a machine learning technique in which knowledge learned from one task is reused for another related task.

In computer vision, a CNN pretrained on a large dataset such as ImageNet learns general visual features including:

- Edges
- Textures
- Shapes
- Patterns
- Higher-level visual structures

These learned features can be transferred to a new dataset.

The general transfer learning process used in this experiment is:

1. Load a pretrained MobileNetV2 model.
2. Remove the original classification head.
3. Freeze the pretrained convolutional base.
4. Add a new classification head for CIFAR-10.
5. Train the new classification head.
6. Unfreeze selected final layers.
7. Fine tune using a smaller learning rate.

11. Dataset

The CIFAR-10 dataset was used for this experiment.

CIFAR-10 contains 60,000 colour images belonging to 10 classes.

Property	Value
Training Images	50,000
Testing Images	10,000
Number of Classes	10
Original Image Size	$32 \times 32 \times 3$
Image Type	RGB

Table 2: CIFAR-10 Dataset Details

The ten classes are:

Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck

12. Dataset Preparation and Preprocessing

The original CIFAR-10 images have a spatial resolution of 32×32 pixels. Since MobileNetV2 was designed for larger input images, the images were resized to 96×96 pixels.

The preprocessing steps were:

1. Load the CIFAR-10 dataset.
2. Convert image values to floating-point representation.
3. Normalize the image values.
4. Resize the images from 32×32 to 96×96 .
5. Apply MobileNetV2 preprocessing.
6. Create TensorFlow dataset pipelines.
7. Shuffle the training data.
8. Use a batch size of 32.

The input shape to the final model is therefore:

$$96 \times 96 \times 3$$

13. MobileNetV2 Architecture

MobileNetV2 is a lightweight CNN architecture designed for efficient image classification and deployment on resource-constrained devices.

It uses depthwise separable convolutions and inverted residual blocks to reduce computational cost.

The pretrained MobileNetV2 model used in this experiment was obtained with ImageNet weights.

The original classification head was removed using:

include_top=False

The convolutional base was then connected to a new classification head.
The resulting architecture is:

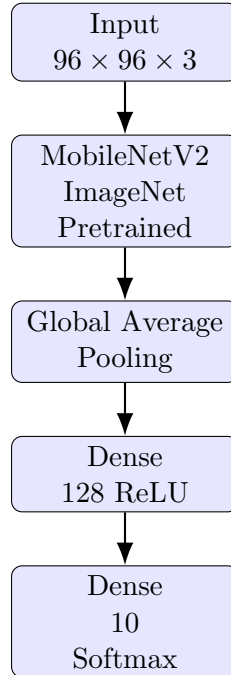


Figure 2: MobileNetV2 Transfer Learning Architecture

14. Model Configuration

The transfer learning model was configured as follows:

Parameter	Value
Base Model	MobileNetV2
Pretrained Weights	ImageNet
Input Size	$96 \times 96 \times 3$
Classification Classes	10
Pooling	Global Average Pooling
Dense Layer	128 neurons
Activation	ReLU
Output Layer	10 neurons
Output Activation	Softmax
Optimizer	Adam
Initial Learning Rate	0.001
Batch Size	32
Initial Training Epochs	10
Fine-Tuning Learning Rate	0.0001
Fine-Tuning Epochs	5

Table 3: Model Configuration

15. Model Training

During the initial transfer learning phase, the MobileNetV2 convolutional base was frozen.

Only the newly added classification layers were trained.

The model was compiled using the Adam optimizer with a learning rate of 0.001.

The loss function used was sparse categorical cross-entropy:

$$L = -\frac{1}{N} \sum_{i=1}^N \log p(y_i|x_i)$$

The model was trained for 10 epochs using a batch size of 32.

The training process was monitored using training and validation accuracy and loss.

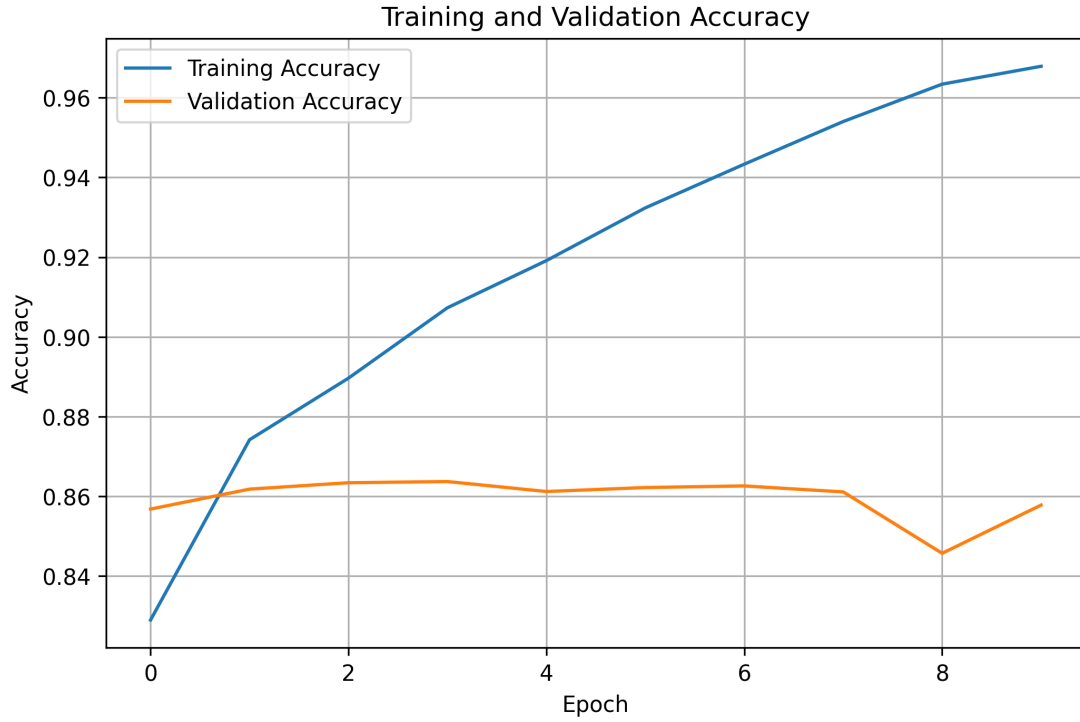


Figure 3: Training and Validation Accuracy

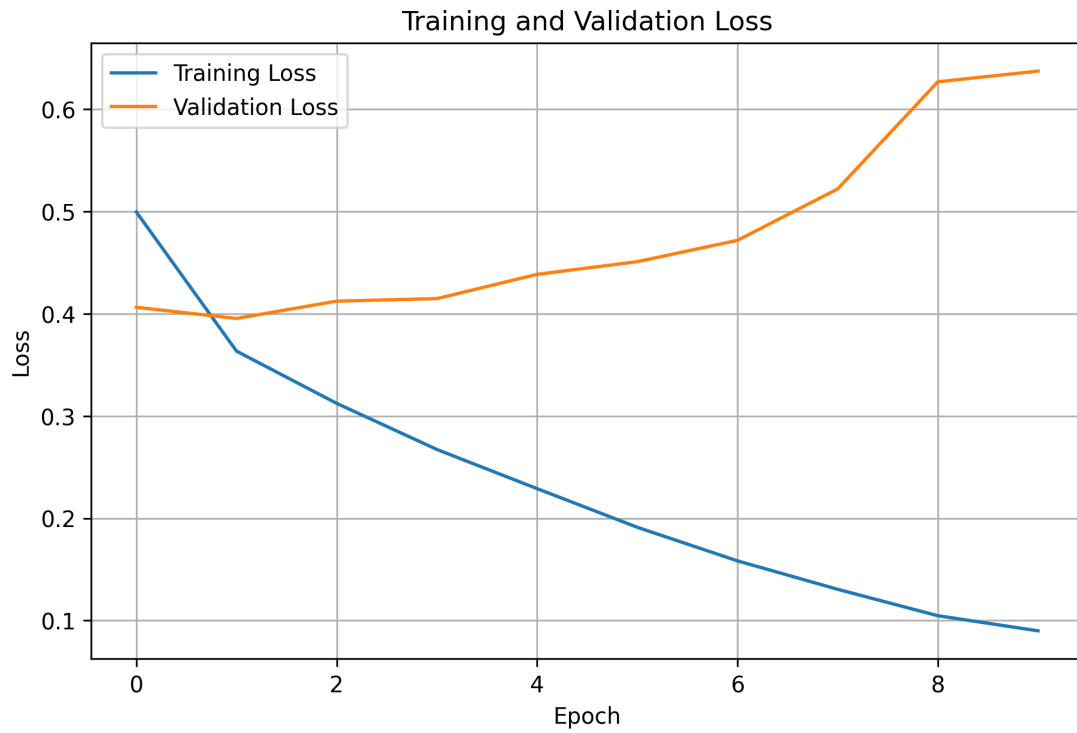


Figure 4: Training and Validation Loss

16. Fine Tuning

After the initial transfer learning stage, selected final layers of the MobileNetV2 convolutional base were unfrozen.

The final 20 layers of the base model were made trainable while the earlier layers remained frozen.

A smaller learning rate of 0.0001 was used during fine tuning to prevent large changes to the pretrained features.

The model was trained for an additional 5 epochs.

The purpose of fine tuning was to allow the higher-level pretrained features to adapt to the CIFAR-10 dataset.

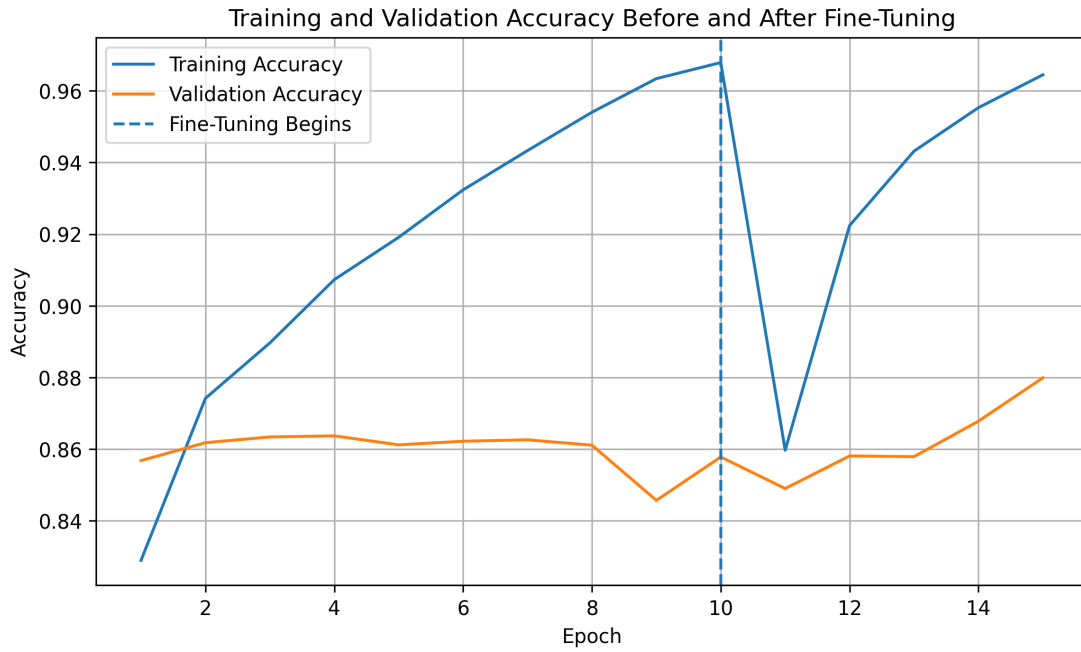


Figure 5: Fine-Tuning Training and Validation Accuracy

17. Model Evaluation

The final fine-tuned model was evaluated using the CIFAR-10 test set.

The following metrics were calculated:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Recall is defined as:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \frac{Precision \cdot Recall}{Precision + Recall}$$

18. Results

18.1 Transfer Learning and Fine Tuning Results

The MobileNetV2 model was first trained with the pretrained convolutional base frozen. The model achieved a testing accuracy of 85.78% before fine tuning.

After unfreezing the selected final layers of the MobileNetV2 convolutional base and fine tuning the model for five additional epochs, the testing accuracy increased to 87.99%.

The improvement in testing accuracy was:

$$87.99 - 85.78 = 2.21$$

percentage points.

The final evaluation metrics and model statistics are shown below.

Metric	Value
Testing Accuracy Before Fine Tuning	85.78%
Testing Accuracy After Fine Tuning	87.99%
Accuracy Improvement	2.21 percentage points
Precision	87.99%
Recall	87.99%
F1-score	87.94%
Total Parameters	2,423,242
Initial Training Time	243.89 seconds
Fine-Tuning Time	186.98 seconds
Total Training Time	430.87 seconds

Table 4: Final Experimental Results

The model achieved an accuracy of 85.78% before fine tuning. After fine tuning the selected layers of MobileNetV2, the testing accuracy increased to 87.99%, resulting in an improvement of 2.21 percentage points.

The final model achieved a precision of 87.99%, recall of 87.99%, and F1-score of 87.94%. The model contained 2,423,242 parameters. The initial training required 243.89 seconds, while fine tuning required an additional 186.98 seconds, resulting in a total training time of 430.87 seconds.

18.2 Effect of Fine Tuning

Stage	Testing Accuracy
Before Fine Tuning	85.78%
After Fine Tuning	87.99%
Improvement	2.21 percentage points

Table 5: Effect of Fine Tuning on Model Accuracy

Fine tuning improved the testing accuracy from 85.78% to 87.99%. This demonstrates that adapting the higher-level pretrained features to the CIFAR-10 dataset improved the classification performance.

18.3 Theoretical Comparison of CNN Architectures

The major CNN architectures studied in this experiment are compared based on their architectural characteristics and major contributions.

The architectures listed below were studied theoretically. LeNet-5, AlexNet, VGG16, GoogleNet and ResNet50 were not independently trained as part of this experiment. Therefore, experimental accuracy and training-time values are not reported for these architectures.

Model	Key Idea	Architecture	Main Advantage
LeNet-5	Convolution and pooling	Shallow CNN	Simple image recognition
AlexNet	Deep CNN and ReLU	Multiple convolutional layers	High classification performance
VGG16	Small 3×3 filters	Deep sequential CNN	Simple and uniform architecture
GoogleNet	Inception modules	Parallel convolution paths	Multi-scale feature extraction
ResNet50	Residual learning	Residual/shortcut connections	Efficient training of deep networks
MobileNetV2	Depthwise separable convolution	Lightweight CNN	Efficient transfer learning

Table 6: Theoretical Comparison of CNN Architectures

19. Confusion Matrix and Misclassification Analysis

The confusion matrix provides a class-wise representation of the predictions made by the final fine-tuned MobileNetV2 model.

Diagonal elements represent correctly classified samples, while off-diagonal elements represent misclassified samples.

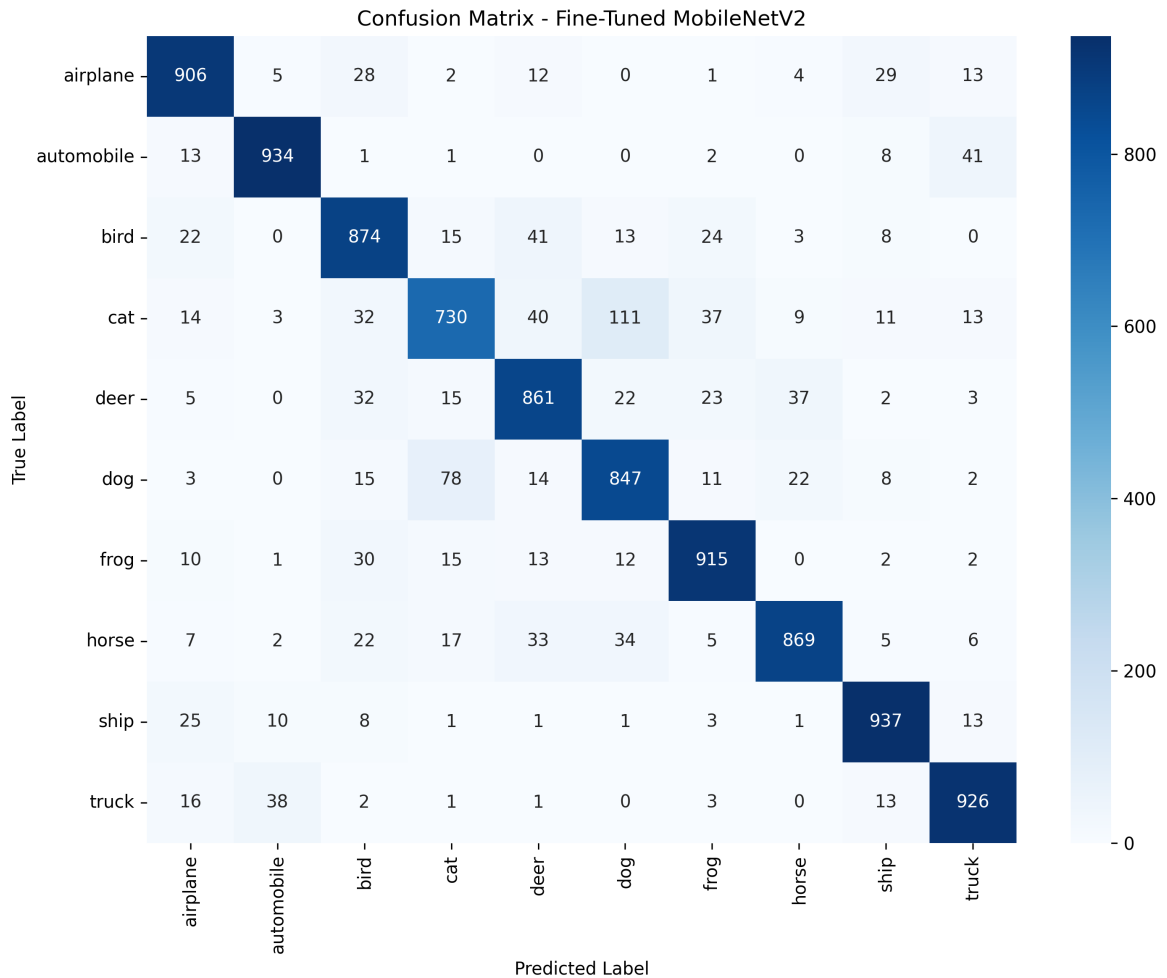


Figure 6: Confusion Matrix of Fine-Tuned MobileNetV2

The confusion matrix can be used to identify classes that are more difficult to distinguish. CIFAR-10 contains visually similar classes, which can result in confusion between certain categories.

19.1 Misclassified Images

Sample misclassified images were visualized to understand the errors made by the model.

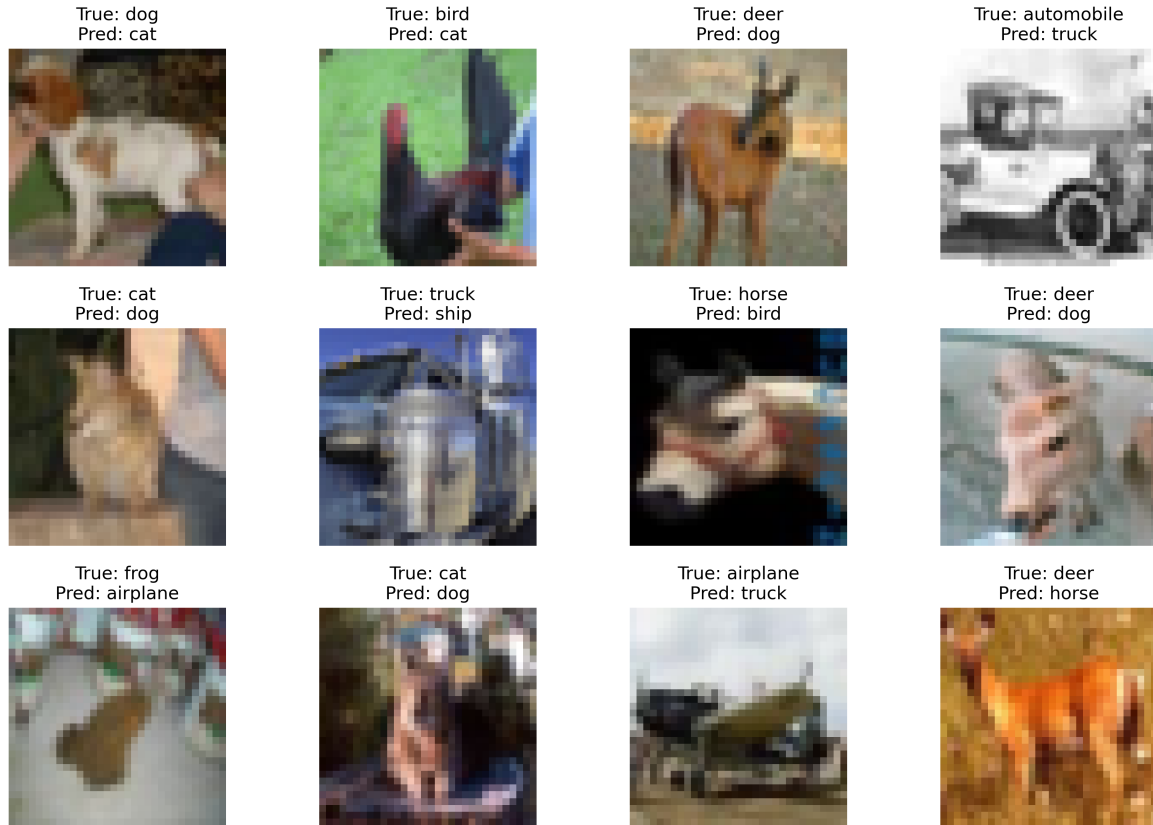


Figure 7: Sample Misclassified Images

Misclassification analysis provides additional insight into cases where the visual features of different classes overlap.

20. Discussion

20.1 Why is AlexNet considered a breakthrough in deep learning?

AlexNet was a major breakthrough because it demonstrated the effectiveness of deep convolutional neural networks for large-scale image classification. It introduced the practical use of ReLU activation functions, GPU-based training, dropout and data augmentation.

20.2 Why does VGG16 use only 3×3 convolution filters?

VGG16 primarily uses small 3×3 convolution filters. Stacking multiple small filters allows the network to obtain a larger effective receptive field while introducing additional nonlinearities between convolutional layers. This provides a simple and uniform architecture.

20.3 What are the advantages of the Inception module?

The Inception module applies different convolution operations and pooling operations in parallel. This enables the network to capture features at different spatial scales while efficiently combining their outputs.

20.4 What is the purpose of residual learning?

Residual learning allows a network to learn a residual mapping instead of directly learning the complete transformation. A residual block can be represented as:

$$y = F(x) + x$$

The shortcut connection helps gradients propagate through deep networks and makes the training of very deep architectures easier.

20.5 Differentiate between LeNet and ResNet.

LeNet is a relatively shallow CNN designed for simple image classification tasks such as handwritten digit recognition. ResNet is a much deeper architecture that uses residual connections to overcome optimization difficulties associated with very deep networks.

20.6 What is Transfer Learning?

Transfer learning is a technique in which knowledge learned by a model on a large source dataset is reused for a related target task. In this experiment, MobileNetV2 pretrained on ImageNet is adapted for CIFAR-10 classification.

20.7 Why is fine tuning required?

Fine tuning allows selected pretrained layers to adapt their learned features to the target dataset. The earlier layers generally capture generic visual features, while higher-level layers can benefit from adaptation to the specific target classes.

20.8 Differentiate between dilated convolution and transpose convolution.

Dilated convolution introduces gaps between the convolutional filter elements, increasing the receptive field without proportionally increasing the number of parameters.

Transpose convolution is commonly used for increasing spatial resolution and is therefore useful in tasks such as image generation and semantic segmentation.

20.9 Why do pretrained models usually converge faster?

Pretrained models already contain useful feature representations learned from a large dataset. Therefore, the model does not need to learn basic visual features entirely from the target dataset, allowing the optimization process to start from a more informative parameter configuration.

20.10 Compare the computational complexity of LeNet and ResNet.

LeNet is a relatively small and shallow network with considerably lower computational requirements. ResNet contains many more layers and parameters and therefore requires substantially greater computational resources. However, residual connections allow ResNet to train effectively despite its increased depth.

21. Additional Exercises

21.1 Implement Transfer Learning using VGG16

Transfer learning can similarly be performed using VGG16 pretrained on ImageNet. The original classification head can be removed and a new classification layer can be added for the target dataset.

21.2 Implement Transfer Learning using ResNet50

ResNet50 can be used as another pretrained CNN by loading ImageNet weights, removing its original classification head and adding a new classification head suitable for the target dataset.

21.3 Compare Adam and SGD Optimizers

Adam and SGD can be compared by training equivalent models using the same dataset, batch size and number of epochs. Accuracy, loss and training time can then be compared.

21.4 Compare Frozen and Fine-Tuned Layers

The results of the initial transfer learning stage and the fine-tuned stage can be compared to determine the effect of adapting pretrained features to the target dataset.

For the MobileNetV2 implementation in this experiment, the testing accuracy increased from 85.78% to 87.99% after fine tuning.

21.5 Evaluate Models using Standard Metrics

Image classification models can be evaluated using Accuracy, Precision, Recall, F1-score and a Confusion Matrix.

The final MobileNetV2 model achieved:

- Precision: 87.99%
- Recall: 87.99%
- F1-score: 87.94%

21.6 Compare Different CNN Architectures

CNN architectures can be compared based on their depth, architectural design, parameter count, computational requirements and classification performance. The architectures studied in this experiment include LeNet-5, AlexNet, VGG16, GoogleNet, ResNet50 and MobileNetV2.

22. Conclusion

In this experiment, the evolution of important CNN architectures was studied, including LeNet-5, AlexNet, VGG16, GoogleNet and ResNet50.

Transfer learning was implemented using MobileNetV2 pretrained on ImageNet for CIFAR-10 image classification.

The initial transfer learning model achieved a testing accuracy of 85.78%. After fine tuning the selected final layers of MobileNetV2, the testing accuracy increased to 87.99%.

Therefore, fine tuning resulted in an improvement of 2.21 percentage points.

The final model achieved a precision of 87.99%, recall of 87.99% and F1-score of 87.94%. The model contained 2,423,242 parameters and the total reported training time was 430.87 seconds.

The experiment demonstrates that pretrained CNN models can provide useful feature representations for new image classification tasks. Fine tuning allows these representations to be adapted to the target dataset and can further improve classification performance.

Source Code

The complete implementation, including dataset preparation, transfer learning, model training, fine tuning, evaluation and visualization, is available in the GitHub repository.

GitHub Repository:

<https://github.com/mokshithreddynarapureddy/DeepLearningLab/tree/main/Experiment-4>